

1 (a)	$g = GM/R^2$	C1	
	$M = 9.81 \times (6.38 \times 10^6)^2 / 6.67 \times 10^{-11}$	M1	
	$= 5.99 \times 10^{24} \text{ kg}$	A0	[2]
	(allow 2 marks if $g = 9.8 \text{ N kg}^{-1}$ used, 1 mark if $g = 10 \text{ N kg}^{-1}$ used)		
(b) (i)	$T = 24 \text{ hours}$	C1	
	$\omega = 2\pi / (24 \times 3600) \text{ or } 2\pi / T$	C1	
	$= 7.27 \times 10^{-5} \text{ rad s}^{-1}$	A1	[3]
	(ii) $m r \omega^2 = GMm / r^2$	C1	
	$r^3 = 7.55 \times 10^{22}$	C1	
	$r = 4.23 \times 10^7 \text{ m}$	A1	[3]